Student Name: Stephen Komolafe ***Student Number: 21336975***

Continuous Assessment Part 1 - *Lucene Search Engine*

Abstract

A Lucene-based information retrieval (IR) system for the Cranfield Collection is designed and evaluated. The system uses Apache Lucene to index approximately 1400 scientific abstracts and evaluates two ranking models: the Vector Space Model (VSM) and BM25, with consistent preprocessing by the EnglishAnalyzer. Both models were tested on all 225 queries using TREC-compatible relevance judgements. TREC Eval metrics were used to assess performance, including Mean Average Precision (MAP), Rprec, bpref, and top-rank precision. The results show that BM25 slightly outperforms VSM in terms of ranking effectiveness, indicating that it is suitable for term-weighted document ranking on this dataset. CranfieldInteractive conducted interactive testing to confirm the system's correctness and usability.

1. Introduction

The purpose of information retrieval systems is to efficiently locate relevant documents within large text collections. In this project, Apache Lucene, a Java-based, modular library for creating full-text search applications, is applied to the Cranfield Collection, which comprises 225 test questions with ranked relevance judgements and 1400 scientific abstracts. The primary objectives were to index the Cranfield Collection using Lucene, implement and evaluate two similarity models, choose an appropriate analyser for pre-processing, evaluate retrieval effectiveness using TREC Eval metrics, and provide a means for end-users to search through these queries in real-time.

2. Methodology

The system was developed on a Linux Ubuntu virtual machine using OpenJDK 24 and Apache Maven 3.9.9 for dependency management. Lucene libraries used include lucene-core, lucene-analyzers-common, and lucene-queryparser (version 8.6.3). The main indexing entry point was defined as ie.tcd.komolafs.CranfieldIndexer.

The EnglishAnalyzer was chosen for content preprocessing, as it tokenises text, converts terms to lowercase, eliminates stop words, and then uses Porter stemming to reduce words to their root forms. Allowing for variations in words such as "compute", "computing", and "computation" to be treated consistently, thus improving recall without reducing precision.

The system indexes all Cranfield documents and applies the same analyser to queries for consistency. The index is stored locally, and search results are written in a TREC-compatible format, for example, results-english-bm25.txt. Two similarity models were compared. The Vector Space Model (VSM), implemented as ClassicSimilarity, ranks documents based on TF-IDF weighting, while BM25Similarity is a probabilistic model accounting for term frequency, document length, and query term weighting.

An additional java file, CranfieldInteractive, was created to allow real-time query input by users from the terminal. This tool loads the existing Lucene index, applies the chosen analyser and similarity model, and retrieves the top-ranked documents for any user query. With this addition it further shows the usability and correctness of the system beyond batch evaluation.

```
vboxuser@inforet:~/CA1/cran$ java -cp target/cran-1.2.jar ie.tcd.komolafs.CranfieldInteractive index english bm25
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::invokeCleaner has been called by org.apache.lucene.store.MMapDirectory (file:/home/vboxuser/CA1/cran/target/cran-1.2.jar)
WARNING: Please consider reporting this to the maintainers of class org.apache.lucene.store.MMapDirectory
WARNING: sun.misc.Unsafe::invokeCleaner will be removed in a future release
Cranfield Search
Type a query (\q to quit)
----------------------------------
>>> maximum

Top 10 results:
 1. DocID=1344 | Score=2.1889 | Title=atmospheric entries with vehicle lift-drag ratio modulated to limit deceleration and rate of deceleration vehicles with maximum lift-drag ratio of
 0. 5.
 2. DocID=1291 | Score=2.1444 | Title=atmosphere entries with spacecraft lift-drag ratios modulated to limit decelerations .
 3. DocID=1347 | Score=2.0151 | Title=approximate analysis of atmospheric entry corridors and angles .
 4. DocID=1177 | Score=1.9262 | Title=effects of rapid heating on strength of airframe components .
 5. DocID=156 | Score=1.9085 | Title=the effect of shallow water on wave resistance .
 6. DocID=1217 | Score=1.8791 | Title=application of inequality constraints to variational problems of lifting re-entry .
 7. DocID=484 | Score=1.8713 | Title=the influence of two-dimensional stream shear for airfoil maximum lift .
 8. DocID=1252 | Score=1.8685 | Title=on the approach to chemical and vibrational equilibrium behind a strong normal shock wave .
 9. DocID=32 | Score=1.7776 | Title=the dynamic motion of a missile descending through the atmosphere .
10. DocID=218 | Score=1.7614 | Title=intensity, scale and spectra of turbulence in mixing region of free subsonic jet .

>>>
```

3. Evaluation

TREC Eval and the relevance judgements from the Cranqrel file were used for evaluation. Since the original file contained formatting inconsistencies that prevented proper evaluation, it was converted into a TREC-compatible format (cranqrel.trec). All 225 queries from the Cranfield Collection were evaluated against both the BM25 and VSM configurations, and performance metrics were computed to compare retrieval effectiveness, as shown in the table below.

Metric	BM25	VSM
MAP (Mean Average Precision)	0.3053	0.3022
Rprec (Precision of the first R relevant docs)	0.3082	0.2934
Bpref (Binary Preference)	0.9523	0.9516
Iprec_at_recall_0.00 (Interpolated precision)	0.5798	0.5773
Iprec_at_recall_0.50	0.3395	0.3198
Iprec_at_recall_1.00	0.1137	0.1179
P_5 (Precision of the 5 first docs)	0.3191	0.3200
P_10 (Precision of the 10 first docs)	0.2333	0.2338
P_100 (Precision of the 100 first docs)	0.0495	0.0504
P_1000 (Precision of the 1000 first docs)	0.0068	0.0068

(Table. Performance metrics obtained from evaluating queries for each similarity model)

The BM25 model achieved a slightly higher mean average precision than VSM, at 0.3053 versus 0.3022. Rprec and bpref were also higher for BM25, with values of 0.3082 and 0.9523, compared to 0.2934 and 0.9516 for VSM. Precision at top ranks demonstrates that both models retrieve relevant documents among the highest-ranked results, with P_5 being 0.3191 for BM25 and 0.3200 for VSM. For both models, iprec gradually declines with increasing recall, indicating that both approaches rank the most relevant documents at the top of the list.

The performance of both BM25 and VSM on the Cranfield Collection was satisfactory. BM25 has a slight advantage due to how it normalises term frequency and document length, making it more robust across documents of varying sizes. The stemming and stop-word removal features in EnglishAnalyzer help to reduce noise and improve term matching. Interactive testing using the CranfieldInteractive tool confirmed that queries are correctly interpreted and that the top-ranked results correspond to relevant documents in the dataset.

4. Conclusion

A Lucene-based search engine was successfully implemented to index and query the Cranfield Collection using both BM25 and VSM. Evaluation on all 225 queries showcased that BM25 provides a slight advantage in retrieval effectiveness, achieving a MAP of 0.3053 compared to 0.3022 for VSM, along with higher Rprec and bpref scores. Both models retrieve relevant documents effectively, especially at top ranks, while the EnglishAnalyzer's stemming and stop-word removal improve term matching and reduce noise. The addition of CranfieldInteractive further confirmed that the system interprets queries correctly and retrieves appropriate top-ranked results.

Bibliography

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